

DATA SCIENCE PROJECT

Bike Sharing Demand Prediction

Objective: Predict the hourly number of bike rentals using temporal, calendar, and weather-related features

 **Presentation:** Heike Thöricht

 **Bremen**, 23. August 2026

Agenda



The Challenge

Why demand forecasting matters



Our Approach

Data-driven solution



Model Performance

Proven accuracy



Key Insights

What drives demand



Implementation & Next Steps

How to use it

Executive Summary — Ready for Go-Live



Prediction Accuracy

89.1% of demand variance explained ($R^2 = 0.891$) — strong, validated accuracy for reliable planning



Forecast Error

Average forecast error: 33 bikes per hour (MAE = 33.0) — realistic margin confirmed through time-series cross validation

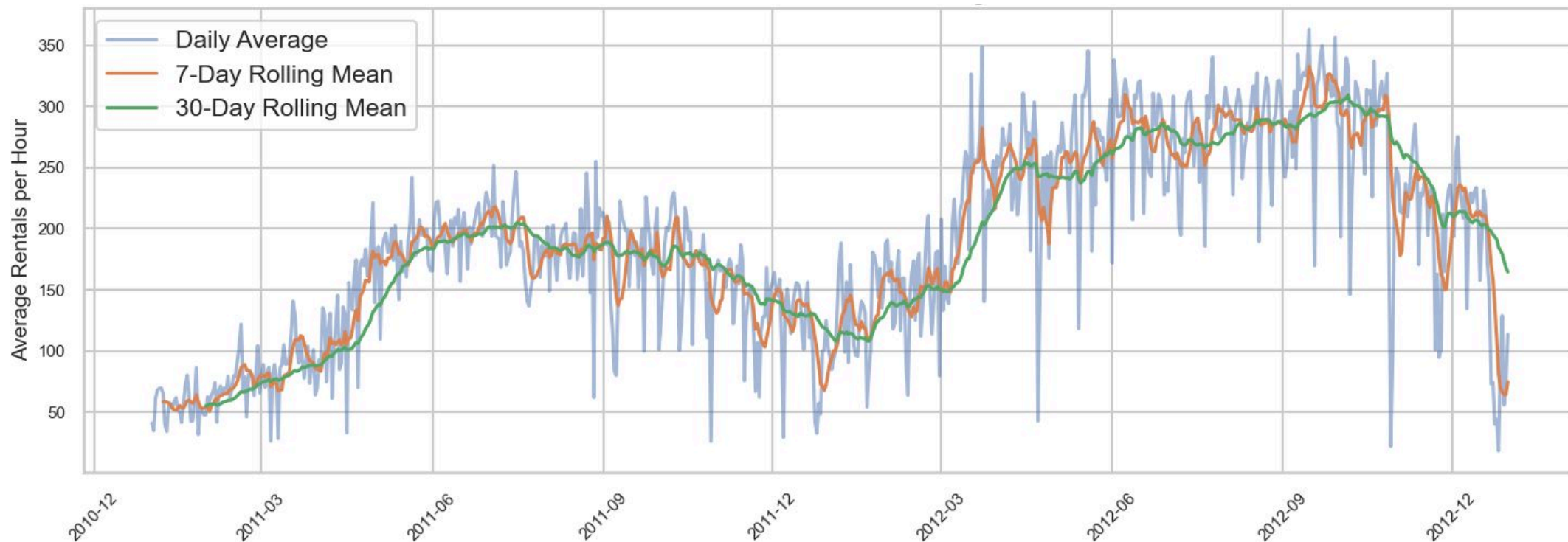


Key Driver

Historical demand patterns (lag_1: 30.7%) and time-of-day (14.1%) drive predictions — highly actionable

The model delivers strong, cross-validated accuracy using historical demand patterns and temporal features — production-ready to support your planning processes with realistic, trustworthy forecasts.

The Rise of Bike Sharing: Emerging Business Opportunities



The Challenge: Why Demand Forecasting Matters

Our Data Foundation

Real-world hourly rental data from the Capital Bikeshare system — a proven operational dataset that mirrors your planning environment.

- **17,379** hourly observations from real bike-sharing operations
- **2 years** of historical data (2011–2012)
- Complete data quality — no missing values

The Business Problem

Without reliable demand forecasts, your planning division faces avoidable inefficiencies and operational risk.

- Unpredictable demand creates **logistics and staffing challenges**
- Core question: **How many bikes are needed each hour?**
- Key insight: **Historical demand patterns (30.7%) + time-of-day (14.1%)** are the strongest predictors
- Goal: Accurate hourly forecasts based on **demand momentum and temporal patterns**

Key Insight: Demand Patterns Drive Planning



Commuter Peaks

Working days show demand spikes at **8 AM** and **5–6 PM** — align bike availability with commute peaks.



Leisure Patterns

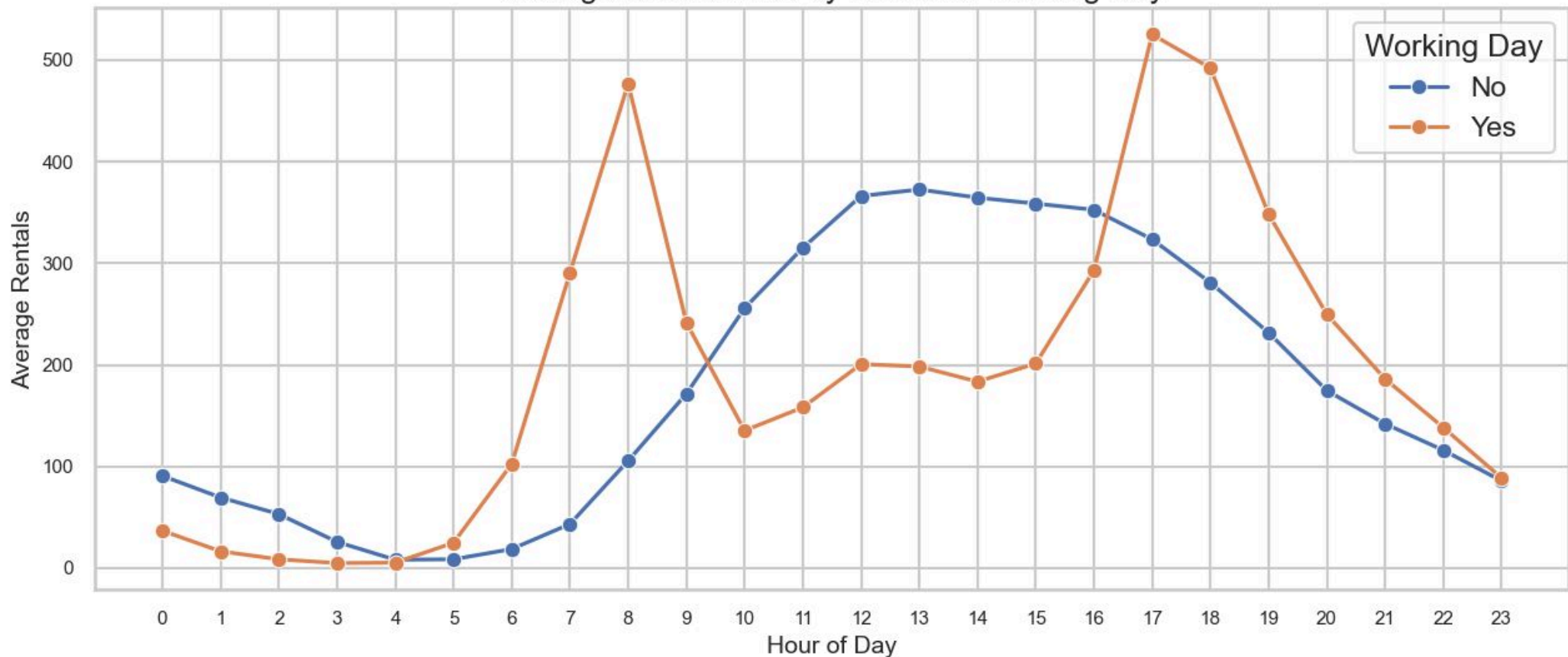
Non-working days show midday peaks — different staffing and bike distribution needed



Actionable

Hour-of-day is the **key planning signal** — enables precise hourly planning

Average Bike Rentals by Hour and Working Day



These patterns are consistent and predictable — your planning can rely on them.

Model Performance: Proven Accuracy

XGBoost delivered the strongest performance in time-series cross-validation, explaining 89.1% of the variation in hourly bike demand ($R^2 = 0.891$).

Model	Feature Set	R^2	MAE	RMSE
Linear Regression	V01	0.876	39.7 bikes	56.5
Random Forest	V01	0.861	36.2 bikes	57.2
 XGBoost (Optuna CV)	V01	0.891	33.0 bikes	50.6

Winner: XGBoost

Strong explanatory performance:
 $R^2 = 0.891$ — explains nearly 9 out of 10 demand variations under rigorous time-series cross validation, a solid and reliable result for production deployment

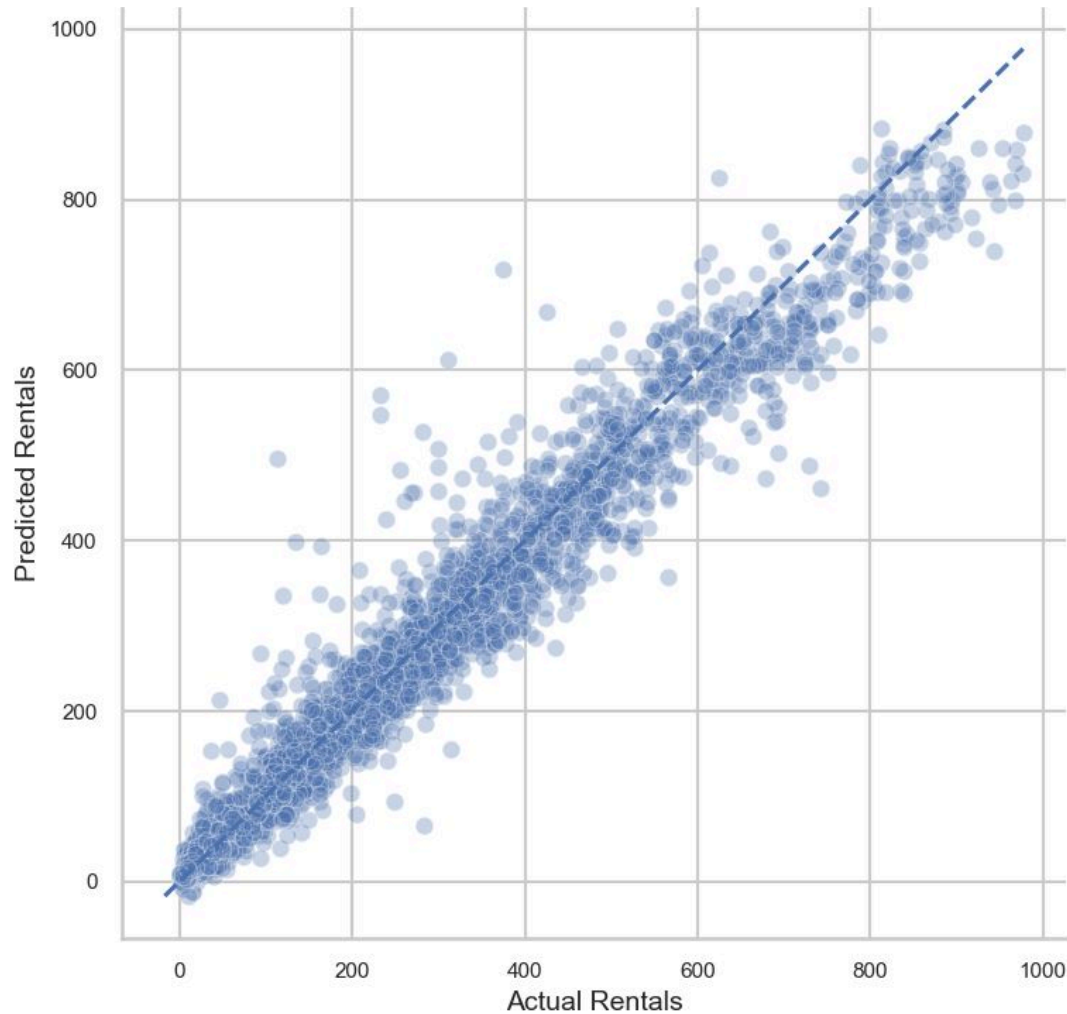
Average forecast error: 33 bikes per hour

Average prediction error of just **33 rentals** — a reasonable margin for operational planning and fleet distribution under real-world conditions

Production-Ready

Tested on **real time-series data** — proven to generalise beyond training conditions and work in live scenarios with consistent, dependable accuracy

Validation: Model Works in Real Scenarios



✓ Overall fit

Predictions closely track actual demand across the evaluation period.

⚠ Peak-hour risk

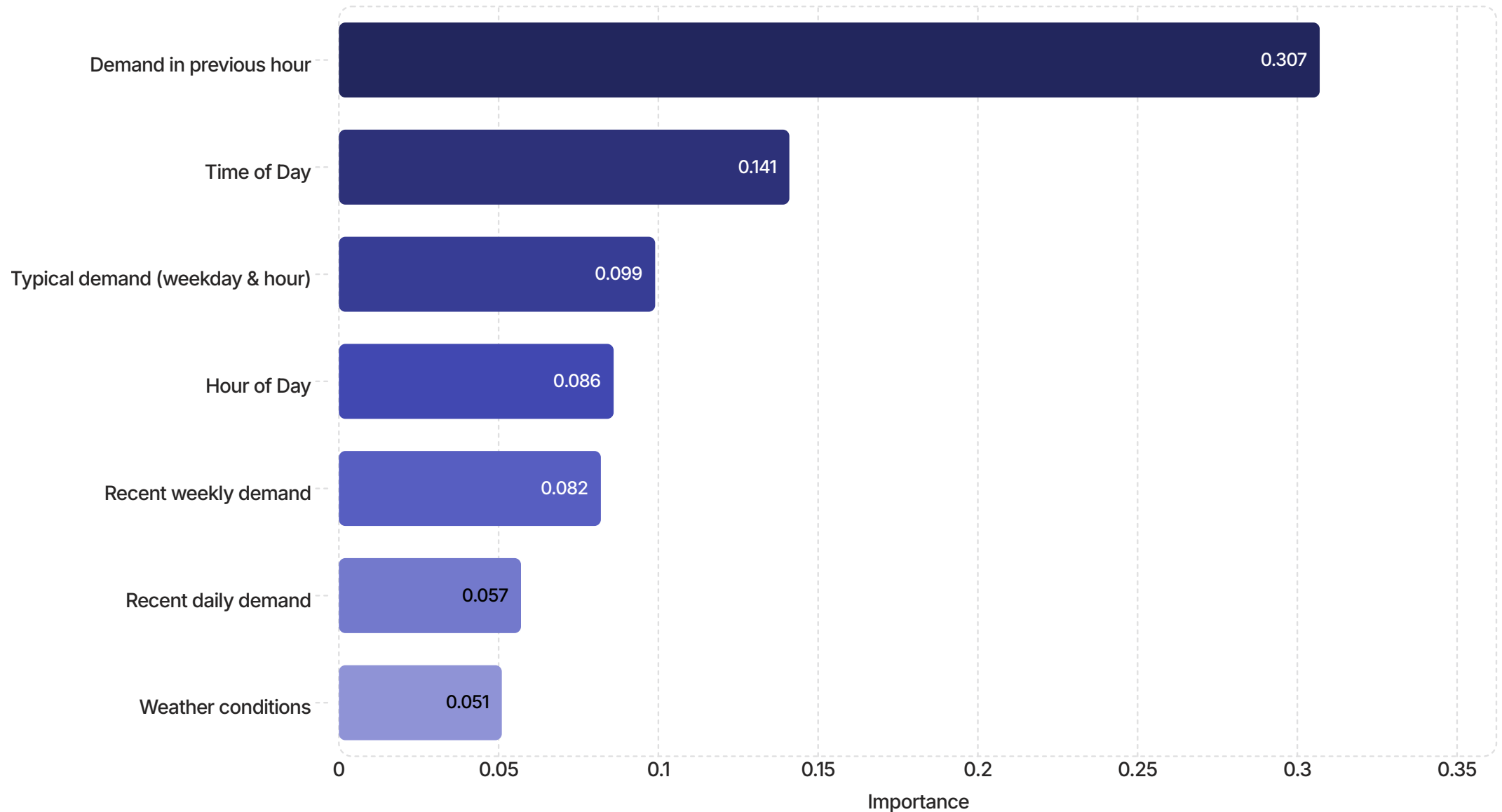
Forecast deviations are larger during demand spikes; peak periods require operational buffer capacity

🎯 Ready for Operations

Use forecasts for daily planning, supported by weekly error monitoring and a defined threshold such as $MAE < 50$ bikes.

What Drives Demand: The Key Factors

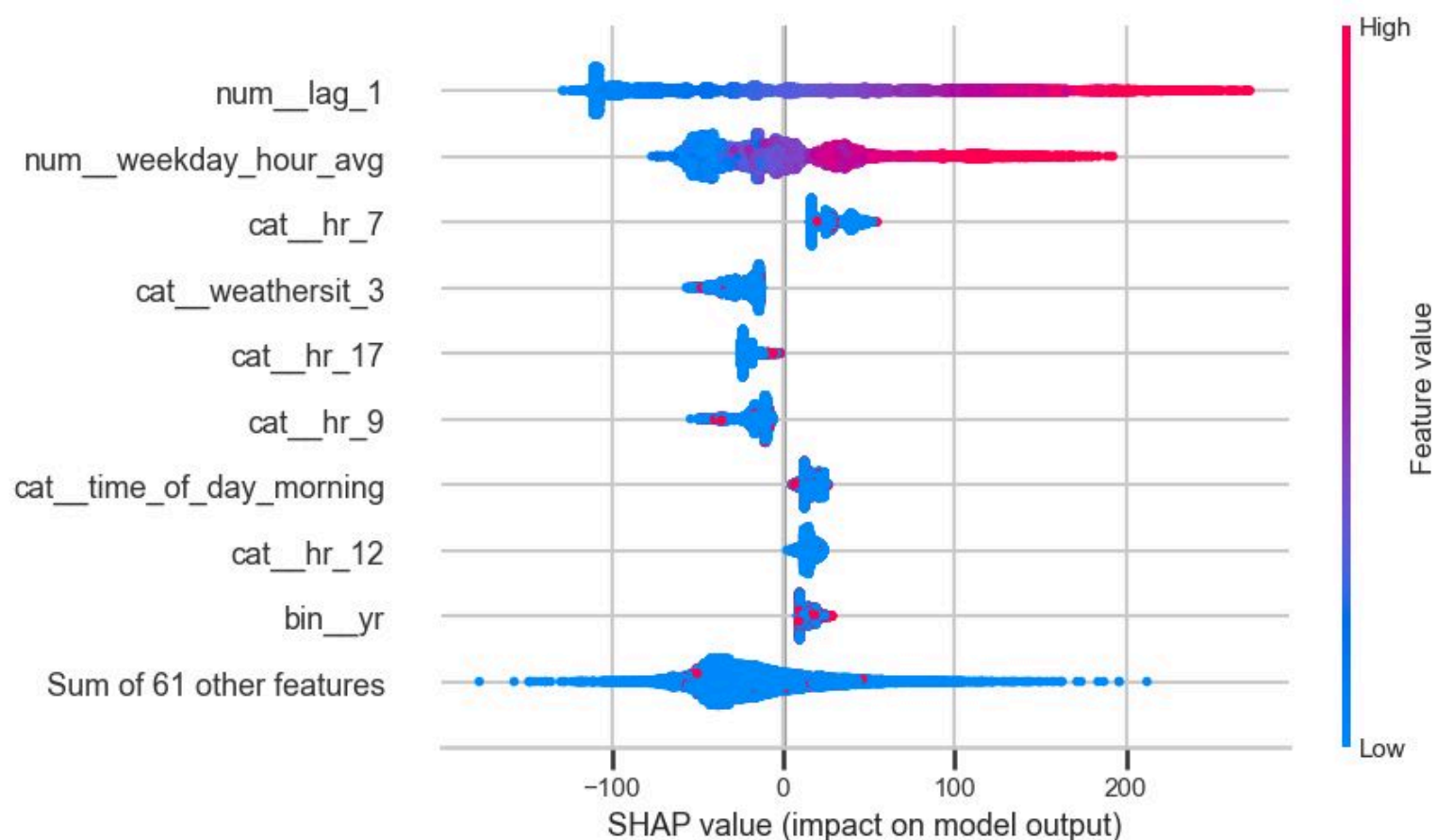
The model's top predictors reveal a clear pattern: recent behavior and time of day dominate, with weather as a fine-tuning signal.



📌 The previous hour's demand alone explains **30.7%** of the forecast — nearly twice the influence of any other single factor.

SHAP Summary Plot

SHAP validates the feature-importance results: recent demand is the strongest signal, historical weekday-hour patterns provide the second key input, and time and weather provides a smaller adjustment to the forecast.



How the Pilot Works

A simple operating model for next-hour demand forecasts.



Refresh the Forecast

EVERY HOUR

Update the next-hour forecast using the latest rental data.



Use It in Operations

EVERY DAY

Support bike availability, rebalancing and staffing decisions.



Monitor and Improve

EVERY WEEK

Review overall forecast error and peak-hour performance.



Model Updates

Retrain quarterly or earlier if forecast quality declines.



Pilot Success Target

Average forecast error below **50 bikes per hour**.

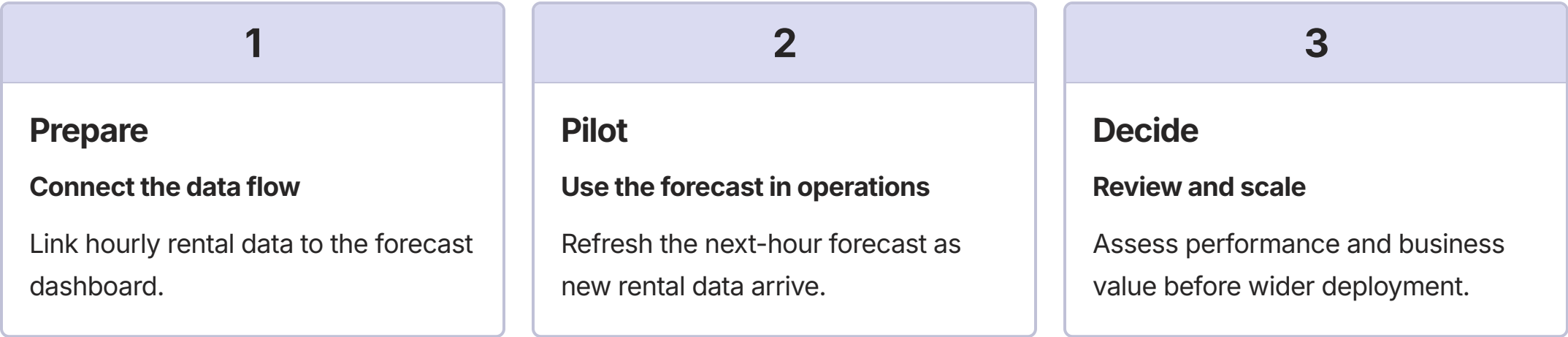


Owners

Operations · Data & Analytics · IT

Next Step: Launch a Controlled Pilot

Recommendation: Start a 4-week pilot using hourly forecasts to support bike availability, rebalancing and staffing decisions.



Pilot success criteria

- Average forecast error remains below **50 bikes per hour**
- Peak-hour performance is monitored and buffers are defined
- Operations confirms that the forecast improves planning decisions

Decision today: Agree pilot scope, nominate owners and confirm the start date.

Week 1: Dashboard and data review · Week 2: Technical integration check · Weeks 3–6: Controlled pilot